Tairan Song

songtai1@msu.edu | tairan_song@outlook.com | (+1) 734-819-0577 | tairan-song.github.io

Research Profile

Ph.D. researcher focused on medical image analysis, with interests in deep learning, and quantitative imaging.

Technical Skills

Programming: Python, R, SQL
Libraries & Tools: PyTorch, scikit-learn, CatBoost, Git, SLURM
ML & Imaging: Graph Representation Learning, Transformers, Metric Learning, Image Analysis, Image Segmentation, Radiomics

Research & Technical Experience

Michigan State University *East Lansing, MI*
Medical Imaging and Data Integration Lab | Graduate Researcher

Supervoxel Graph Representation Learning for Renal Mass Classification Jan 2026 – Present
Advisor: Dr. Adam Alessio

- Built 3D tumor graphs with center-growing supervoxels, using 23 local CT descriptors per node and six-neighbor spatial edges to preserve intratumoral tissue organization.
- Developed a graph Transformer with learnable shortest-path attention biases, attention pooling, and triplet loss to learn topology-aware tumor-level embeddings.
- Benchmarked embeddings with CatBoost and patient-grouped nested five-fold validation: mean ROC-AUC 0.77 ± 0.01 versus 0.75 ± 0.03 without structural bias and 0.67 ± 0.08 for whole-tumor radiomics.

Quantitative CT Analysis for Renal Mass Characterization Jan 2025 – Dec 2025
Advisor: Dr. Adam Alessio

- Built a 3D CT processing pipeline for 360 examinations from 312 patients, combining TotalSegmentator, a KiTS23-trained nnU-Net, regional radiomics, and fractal-dimension features.
- Evaluated contrast-reference features from kidney and aorta using CatBoost and patient-grouped nested cross-validation; achieved mean ROC-AUC 0.78 ± 0.05 for benign-versus-malignant classification.

Zhejiang University *Hangzhou, China | Sep 2022 – Mar 2023*
Research Assistant | Computational Cancer Genomics
Advisor: Dr. Ning Shen

- Implemented and validated RESA-jLR, combining logistic regression on variant-quality metrics and sequence-context features to identify candidate somatic SNVs from scRNA-seq; benchmarked against existing mutation-calling methods.

Publications & Manuscripts

- **Song, T.**, Huang, Z., Noyes, S. L., et al. "Supervoxel Graph Representation Learning for Renal Mass Classification from CT." *SPIE Medical Imaging 2027* (accepted).
- Munavar Ali, M. A., **Song, T.**, Noyes, S. L., et al. "Histologic Grade Prediction in High-Risk Renal Masses from Multi-Region CT Analysis." 2026 Leaders on the Horizon Program, *Applied Radiology* (accepted).
- Zhang, T., Jia, H., **Song, T.**, et al. "De novo identification of expressed cancer somatic mutations from single-cell RNA sequencing data." *Genome Medicine* 15, 115 (2023).

Conference Presentations

- **SPIE Medical Imaging 2027**: "Supervoxel Graph Representation Learning for Renal Mass Classification from CT." Oral presentation (scheduled).
- **Radiological Society of North America (RSNA) 2026 Annual Meeting**: "Core-to-Margin Radiomic Profiling for Predicting Kidney Cancer Grade from Variable-Phase CT." Scientific poster (scheduled).

Education

Michigan State University, East Lansing, MI, USA
Ph.D. Student, Computational Mathematics, Science and Engineering; GPA: 3.8/4.0 Aug 2023 – Present
M.S., Computational Mathematics, Science and Engineering; GPA: 3.8/4.0 Aug 2023 – Aug 2026

King's College London, London, UK
M.Sc., Applied Statistical Modelling and Health Informatics (Distinction) Sep 2021 – Sep 2022

University of Liverpool, Liverpool, UK
B.Sc., Mathematics and Statistics (Upper Second-Class Honours) Sep 2018 – Jun 2021

Additional Experience

Teaching Assistant, CMSE 202: Computational Modeling and Data Analysis II, MSU Jan – May 2025
Financial Data Analytics Intern, China Galaxy Securities Co., Ltd., Shanghai Jul – Sep 2021

Awards & Honors

NSF NRT-IMPACTS Training Support, Michigan State University Fall 2023